

Partial Dependency

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What is Partial Dependency?

Formal Definition: A Partial Dependency exists when a non-prime attribute is functionally dependent on only a part of a composite Candidate Key, instead of the entire key.

Example:

Candidate Key: (StudentID, CourseID)

Suppose: StudentID $\rightarrow$ StudentName

StudentName does not need CourseID. Therefore, StudentName is only partially dependent.

Why is this a Problem?

Because StudentName will be repeated for every course taken by the same student.

StudentID	CourseID	StudentName
101	DB101	Aadya
101	OS201	Aadya
101	CN301	Aadya

The value "Aadya" is stored repeatedly. This causes: Redundancy, Update anomalies, Storage waste.

Important Rules

- Partial Dependency can only occur when there is a Composite Candidate Key.
- Partial Dependency always involves a: Non-prime Attribute.

Detecting Partial Dependency

1. **Step 1:** Find the Candidate Key.
2. **Step 2:** Check if it is Composite. If not, stop. No Partial Dependency exists.
3. **Step 3:** Check every Functional Dependency. Ask: Does the dependent attribute depend on the entire key or only part of it?

Removing Partial Dependency (Converting to 2NF)

Partial Dependencies are removed by decomposing the table.

Table 1 (Student): | StudentID | StudentName |

Table 2 (Course): | CourseID | CourseName |

Table 3 (Enrollment): | StudentID | CourseID | Grade |

Now: StudentName depends entirely on StudentID. CourseName depends entirely on CourseID. Grade depends entirely on (StudentID, CourseID). No Partial Dependency remains. This is Second Normal Form (2NF).

Example 4 – Hospital

| PatientID | DoctorID | PatientName | Diagnosis |

FDs: (PatientID, DoctorID) → Diagnosis, PatientID → PatientName

Partial Dependency: PatientID → PatientName